

Segmentación De Clientes Con Análisis RFM

En Power BI Usando DAX

1. Tabla Calculada: RFM Table

Descripción:

Tabla que resumen por cliente las métricas *Recency*, *Frequency* y *Monetary*, base del análisis RFM.

```
RFM Table =  
VAR _LastPurchaseDate =  
    CALCULATE(  
        MAX('Online Retail'[InvoiceDate]),  
        REMOVEFILTERS('Online Retail')  
    )  
VAR _RFMTable =  
    ADDCOLUMNS(  
        VALUES('Online Retail'[CustomerID]),  
        "Recency", DATEDIFF(CALCULATE(MAX('Online Retail'[InvoiceDate])), _LastPurchaseDate, DAY),  
        "Frequency", CALCULATE(DISTINCTCOUNT('Online Retail'[InvoiceNo])),  
        "Monetary", CALCULATE(SUM('Online Retail'[Revenue]))  
    )  
RETURN _RFMTable
```

2. Medidas: Percentiles Para Recency

Descripción:

Medias que calculan los percentiles 33% y 66% de la variable *Recency*.

```
Recency 33 Percentile =
VAR _LastPurchaseDate =
    CALCULATE(
        MAX('Online Retail'[InvoiceDate]),
        REMOVEFILTERS('Online Retail')
    )
VAR _RFMTable =
    ADDCOLUMNS(
        VALUES('Online Retail'[CustomerID]),
        "@Recency", DATEDIFF(CALCULATE(MAX('Online Retail'[InvoiceDate])), _LastPurchaseDate, DAY)
    )
RETURN
    PERCENTILEX.INC(_RFMTable, [@Recency], 0.33)

Recency 66 Percentile =
VAR _LastPurchaseDate =
    CALCULATE(
        MAX('Online Retail'[InvoiceDate]),
        REMOVEFILTERS('Online Retail')
    )
VAR _RFMTable =
    ADDCOLUMNS(
        VALUES('Online Retail'[CustomerID]),
        "@Recency", DATEDIFF(CALCULATE(MAX('Online Retail'[InvoiceDate])), _LastPurchaseDate, DAY)
    )
RETURN
    PERCENTILEX.INC(_RFMTable, [@Recency], 0.66)
```

3. Medidas: Percentiles Para Frequency

Descripción:

Percentiles 33% y 66% de la *frecuencia de compra* por cliente.

```
Frequency 33 Percentile =  
VAR _RFMTable =  
    ADDCOLUMNS(  
        VALUES('Online Retail'[CustomerID]),  
        "@Frequency", CALCULATE(DISTINCTCOUNT('Online Retail'[InvoiceNo]))  
    )  
RETURN  
    PERCENTILEX.INC(_RFMTable, [@Frequency], 0.33)  
  
Frequency 66 Percentile =  
VAR _RFMTable =  
    ADDCOLUMNS(  
        VALUES('Online Retail'[CustomerID]),  
        "@Frequency", CALCULATE(DISTINCTCOUNT('Online Retail'[InvoiceNo]))  
    )  
RETURN  
    PERCENTILEX.INC(_RFMTable, [@Frequency], 0.66)
```

4. Medidas: Percentiles Para Monetary

Descripción:

Percentiles 33% y 66% del *gasto total* por cliente.

```
Monetary 33 Percentile =  
VAR _RFMTable =  
    ADDCOLUMNS(  
        VALUES('Online Retail'[CustomerID]),  
        "@Monetary", CALCULATE(SUM('Online Retail'[Revenue]))  
    )  
RETURN  
    PERCENTILEX.INC(_RFMTable, [@Monetary], 0.33)  
  
Monetary 66 Percentile =  
VAR _RFMTable =  
    ADDCOLUMNS(  
        VALUES('Online Retail'[CustomerID]),  
        "@Monetary", CALCULATE(SUM('Online Retail'[Revenue]))  
    )  
RETURN  
    PERCENTILEX.INC(_RFMTable, [@Monetary], 0.66)
```

5. Columnas Calculadas: Scores

Descripción:

Cálculo de *R*, *F* y *M* Scores en la *RFM Table*.

```
R Score =  
SWITCH(  
    TRUE(),  
    'RFM Table'[Recency] <= [Recency 33 Percentile], 3,  
    'RFM Table'[Recency] <= [Recency 66 Percentile], 2,  
    1  
)  
  
F Score =  
SWITCH(  
    TRUE(),  
    'RFM Table'[Frequency] <= [Frequency 33 Percentile], 1,  
    'RFM Table'[Frequency] <= [Frequency 66 Percentile], 2,  
    3  
)  
  
M Score =  
SWITCH(  
    TRUE(),  
    'RFM Table'[Monetary] <= [Monetary 33 Percentile], 1,  
    'RFM Table'[Monetary] <= [Monetary 66 Percentile], 2,  
    3  
)
```

6. Columna Calculada: RFM Score

Descripción:

Genera una cadena de texto con los tres scores para facilitar la segmentación.

```
RFM Score =  
FORMAT([R Score], "0") & FORMAT([F Score], "0") & FORMAT([M Score], "0")
```

7. Columna Calculada: Segmentación De Clientes

Descripción:

Clasifica a cada cliente en un segmento basado en su combinación *RFM*.

```
Segment =  
SWITCH(  
    TRUE(),  
    [R Score] = 3 && [F Score] = 3 && [M Score] = 3, "VIP Customer",  
    [R Score] = 3 && [F Score] >= 2 && [M Score] >= 2, "Loyal Customer",  
    [R Score] = 3, "New Customer",  
    [R Score] = 2 && [F Score] = 3, "Potential Loyallist",  
    [F Score] = 1 && [M Score] = 1, "At Risk",  
    [R Score] = 1, "Lost Customer",  
    "Promising Customer"  
)
```


8. Resultados Del Código DAX

CustomerID	Recency	Frequency	Monetary	R Score	F Score	M Score	RFM Score	Segment
12346	326	1	77183.60	1	1	3	113	Lost Customer
12347	2	7	4310.00	3	3	3	333	VIP Customer
12348	75	4	1797.24	2	2	3	223	Promising Customer
12349	18	1	1757.55	3	1	3	313	New Customer
12350	311	1	334.40	1	1	1	111	At Risk
12352	36	8	2506.04	2	3	3	233	Potential Loyallist
12353	204	1	89.00	1	1	1	111	At Risk
12354	232	1	1079.40	1	1	2	112	Lost Customer
12355	214	1	459.40	1	1	2	112	Lost Customer
12356	22	3	2811.43	3	2	3	323	Loyal Customer
12357	33	1	6207.67	2	1	3	213	Promising Customer
12358	1	2	1168.06	3	2	3	323	Loyal Customer
12359	57	4	6372.58	2	2	3	223	Promising Customer
12360	52	3	2662.06	2	2	3	223	Promising Customer
12361	288	1	189.90	1	1	1	111	At Risk
12362	3	10	5226.23	3	3	3	333	VIP Customer
12363	109	2	552.00	1	2	2	122	Lost Customer
12364	7	4	1313.10	3	2	3	323	Loyal Customer
12365	292	2	641.38	1	2	2	122	Lost Customer
12367	4	1	168.90	3	1	1	311	New Customer
12370	51	4	3545.69	2	2	3	223	Promising Customer
12371	44	2	1887.96	2	2	3	223	Promising Customer
12372	71	3	1298.04	2	2	3	223	Promising Customer
12373	312	1	364.60	1	1	1	111	At Risk
12374	25	1	742.93	2	1	2	212	Promising Customer
12375	10	2	457.50	3	2	2	322	Loyal Customer
12377	316	2	1628.12	1	2	3	123	Lost Customer
12378	129	1	4008.62	1	1	3	113	Lost Customer
12379	81	2	852.24	2	2	2	222	Promising Customer
12380	21	4	2724.81	3	2	3	323	Loyal Customer
12381	4	5	1845.31	3	3	3	333	VIP Customer
12383	184	5	1850.56	1	3	3	133	Lost Customer
12384	28	2	585.27	2	2	2	222	Promising Customer
12386	338	2	401.90	1	2	2	122	Lost Customer
12388	15	6	2780.66	3	3	3	333	VIP Customer
12390	79	1	549.84	2	1	2	212	Promising Customer
12391	21	1	460.89	3	1	2	312	New Customer
12393	72	4	1582.60	2	2	3	223	Promising Customer
12394	63	2	1272.48	2	2	3	223	Promising Customer
12395	19	12	3018.63	3	3	3	333	VIP Customer
12397	35	2	2409.90	2	2	3	223	Promising Customer

RFM Table (4,338 rows)